

Obstacle avoidance under perturbation: a closed-loop attractor account

Summary

A central problem in systems motor neuroscience is how population activity is organized during rapid, graded feedback corrections that preserve goal-directed reaching when unexpected forces drive the limb toward environmental constraints. We asked whether reward-based learning alone can produce route-specific obstacle avoidance, and whether internal activity and closed-loop dynamics can account for recovery and failure. We trained a proximal policy optimization (PPO) actor-critic for 3 million environment steps. Its feedforward multilayer perceptron actor had two 256-unit hidden layers and mapped the current kinematic state and task geometry to an action. During evaluation, brief lateral force impulses challenged the controller as a point mass traversed a two-obstacle corridor. We analyzed one 256-unit hidden layer using principal component analysis, clustering, kinematic tuning, linear decoding, and dynamical analysis of the policy coupled to the environment. Moderate impulses changed the lateral path by less than 0.1 m without interrupting forward progression. The first two principal components captured 71% of hidden-unit activity variance; trajectories through this space were ordered by movement phase and separated by detour route. At the obstacle band, closed-loop analysis identified one locally stable lateral fixed point per route ($x^*=3.32$ m and 6.71 m). Linearization indicated non-oscillatory local return, with relaxation times of 1.42 s and 1.34 s and obstacle clearances of 0.43 m and 0.46 m. A linear decoder identified perturbation condition from hidden activity after the impulse (accuracy 0.81; chance 0.14), but not before it (0.11). These results link route-structured population activity to local stabilization in this learned controller. They do not establish neural equivalence: the artificial population, point-mass dynamics, and single training run constrain interpretation. Instead, this framework yields testable predictions about population activity and feedback stabilization during perturbed biological reaching.

Additional details

A central question in systems motor neuroscience is how population-level activity relates to feedback control of ongoing movement. During reaching, motor-cortex activity evolves through structured population trajectories [1]. Here, "population" refers to 256 artificial hidden units, not recorded neurons; the comparison is at the level of analytical organization, not neural mechanism. The actor is feedforward and has no recurrent memory, so its hidden activity is a static function of the current observation rather than an autonomous dynamical system. We therefore analyze the closed loop formed by the policy and the physics: the policy maps the current state to an action, the environment updates position and velocity, and the new state feeds back to the policy. The fixed points belong to this policy-environment system, in physical state space, rather than to hidden activity as in recurrent-network analyses [2].

The PPO actor-critic was trained for 3 million environment steps. Its observation comprised velocity and positions of the goal and obstacles relative to the point mass; reward combined goal progress, a speed penalty, collision avoidance, and a success bonus. Evaluation included an unperturbed baseline and transient lateral forces applied for 14 timesteps from step 40. Leftward force magnitudes were 0.40, 0.60, and 0.85; rightward magnitudes were 0.40, 0.65, and 1.80. The magnitudes were not matched across directions, and both detour routes were tested. We analyzed hidden-unit activity with principal component analysis, clustering, kinematic tuning, and linear decoding, and examined the closed-loop response with the perturbation removed. Results are based on one checkpoint, one training seed, and 40 episodes per condition; they do not characterize between-seed variability.

From population activity to movement

The first two principal components captured 71% of hidden-unit variance. Projected population trajectories were ordered by movement phase, shared across perturbation conditions, and separated by detour route; within-route clusters likewise followed time rather than force condition. We tested local stabilization by measuring commanded lateral acceleration at $y = 5$ m with the force removed and forward speed fixed at its plateau. A negative-slope zero crossing means small lateral deviations elicit a command back toward that state. PCA describes population geometry; the closed-loop vector field tests local corrective dynamics.

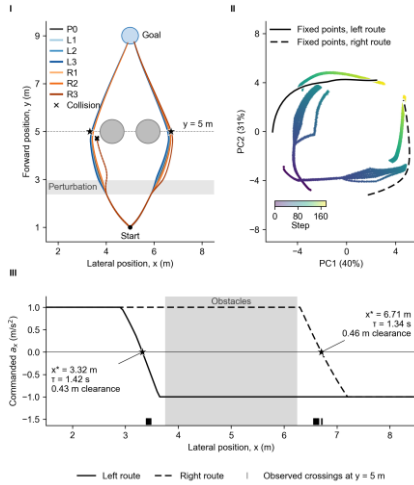


Figure 1. Behaviour, population-state structure, and closed-loop stability. (A) Baseline and perturbed trajectories; collisions occurred only for the strongest rightward impulse on the left route. Stars mark route-specific fixed points at $y = 5$ m. (B) Hidden-unit activity projected onto PC1-PC2 and colored by movement step. (C) Commanded lateral acceleration at $y = 5$ m; the negative-slope zero crossing on each route indicates local stability.

At $y = 5$ m, route-specific fixed points ($x = 3.32$ m and 6.71 m) lay within 0.11 m and 0.09 m of successful crossings and left 0.43 m and 0.46 m of obstacle clearance. Linearization yielded real eigenvalues with magnitudes below one, indicating non-oscillatory local stability. Relaxation times (1.42 s and 1.34 s) characterize this checkpoint, not human response latency; locations also varied across checkpoints. Figure 2 tests whether perturbation condition is decodable from the artificial population.

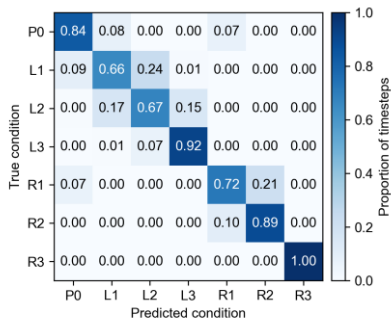


Figure 2. Episode-held-out decoding of perturbation condition from instantaneous hidden-unit activity after the impulse. Rows indicate true conditions and columns decoder predictions; entries are row-normalized proportions. Accuracy was 0.81 (chance 0.14), with most errors between adjacent force magnitudes on the same side.

An episode-held-out multinomial logistic decoder classified perturbation condition from instantaneous activity above chance after the impulse (0.81 ; chance 0.14), but not before it (0.11). This establishes decodability, not policy use. Biological reaching also involves graded force corrections during ongoing movement [3,4], but this model has neither muscles nor neural recordings. It predicts that route and local recovery may covary with neural and muscle population activity during perturbed reaching.

References

- [1] Churchland et al. (2012)
- [2] Sussillo & Barak (2013)
- [3] Pruszynski & Scott (2012)
- [4] Nashed et al. (2014)